1. Identity

Document	STAR-PKG-STAR-CDM-EN · v2.0 · 2026-10-21
Package	star-cdm
Version	0.2.2
Capability	star.ssa.cdm
Maturity	VERIFIED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_cdm

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_cdm-0.2.2-py3-none-any.whl
```

FILE	SHA-256
star_cdm-0.2.2-py3-none-any.whl	c04a36cb93efb0f7e6c5429dcfc01e7e9d1f3b505aae77f06a9119c7ebb2036e
star_cdm.py	101363f1c48f3a735efa1e1ef3670b58eb75ad1d698b174bd3213b0cc8a5b8a2
star_cdm-0.2.2-source.zip	51b99f9ce745c9beb87fc3ce825b7c1921492e0b3c8edc702d788b593c358325

5. What it does, and its limits

Strict parser for CCSDS 508.0-B-1 Conjunction Data Messages in KVN. Standard library only.

XML CDM, CDM 2.0 (508.0-B-2 draft/updates), collision probability computation, any operational use.

- `parse_cdm(text)` → `{"header", "object1", "object2", "warnings"}`: dates as datetimes (CCSDS 502.0 formats only), numbers as floats, units checked against the standard, exactly OBJECT1 then OBJECT2, all obligatory keywords (the set of the standard's example 3.6.2).
- Every malformation raises `CdmFormatError` (unknown keyword, missing obligatory keyword, wrong unit, duplicate, non-numeric, malformed date, wrong object count). Typographic minus signs (U+2212) are normalised and REPORTED.
- `covariance_rtn(obj)` → symmetric 6×6; `consistency_report(cdm)` compares MISS_DISTANCE with the state vectors and RELATIVE_POSITION_R/T/N and flags inconsistencies without editing values.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_cdm_0_2_2,
  author  = {Cavazzini, Andrea},
  title   = {{star-cdm}: Strict CCSDS 508.0-B-1 Conjunction Data Message (KVN) parser with
consistency report},
  version = {0.2.2},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_cdm/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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